High-Fidelity Architectural Framework for Passive Coherent Location Systems: Optimal Signal Processing and Multistatic Integration

The paradigm of Passive Coherent Location (PCL), colloquially referred to as passive radar, has transitioned from a theoretical curiosity to a cornerstone of modern covert electronic surveillance.[1] Unlike conventional active radar systems that require dedicated, high-power transmitters to illuminate targets, PCL systems operate by exploiting non-cooperative illuminators of opportunity (IoO), such as FM radio, Digital Video Broadcasting (DVB-T), Global System for Mobile Communications (GSM), and increasingly, 5G NR networks.[3] The inherent advantages of such a system—stealth, spectral efficiency, and resilience to anti-radiation missiles—are balanced by the significant computational and signal processing challenges required to extract extremely weak target echoes from a sea of high-power interference.[1]

The realization of a "perfect" passive radar necessitates an architectural framework where every processing block is optimized for maximum dynamic range and sensitivity.[1] This report provides an exhaustive technical analysis of the optimal PCL processing pipeline, detail-oriented block-by-block specifications, and the mathematical rigor required for the implementation of a high-performance system.

Architectural Overview and Geometric Constraints

The fundamental geometry of a PCL system is bistatic or multistatic, characterized by the physical separation of the transmitter and the receiver.[4] This separation introduces complex geometric relationships that govern the system's resolution and coverage.[3] The "perfect" system must account for these non-linearities in its initial signal selection and late-stage tracking.[3]

Geometric Resolution Metrics

In a bistatic configuration, the range and Doppler resolution are not constant but depend on the bistatic angle β, which is the angle subtended by the transmitter and receiver at the target.[8] The bistatic range resolution ΔR is fundamentally limited by the bandwidth B of the signal of opportunity and the system geometry:

$$\Delta R = \frac{c}{2B\cos(\beta/2)}$$

As the target approaches the baseline (the direct line between transmitter and receiver), β approaches 180 degrees, causing the range resolution to degrade toward infinity.[8] Similarly, the Doppler resolution Δv is a function of the coherent processing interval (CPI), denoted as T_{CPI}:

$$\Delta v = \frac{\lambda}{2T_{CPI}\cos(\beta/2)}$$

The choice of T_{CPI} represents a critical trade-off: longer intervals provide finer velocity resolution but increase the risk of range and Doppler migration, where a target moves across multiple resolution cells during the integration period.[12]

The Optimal Passive Radar Block Diagram: Interconnects and Signal Flows

The following table delineates the high-level block diagram of a perfect PCL system. Each block represents a critical stage in the signal processing chain, from raw electromagnetic capture to the final display of tracked target trajectories.

Block ID	Block Name	Input Interconnects	Output Interconnects	Optimal Processing Technique
B1	Dual-Channel RF Frontend	EM Waves (Direct & Surveillance)	Digitized I/Q Baseband Streams	Phase-Coherent SDR with High Dynamic Range [4]
B2	Digital Beamforming (DBF)	Multi-element I/Q Streams	Spatially Filtered Surveillance Signal	Adaptive Null-Steering & Direction Finding [2]
B3	Reference Signal	Digitized Reference	Purified/Perfect Reference	Demodulation-Remodulation

	Reconstruction	Stream	Replica	with FEC [15]
B4	Clutter & DPI Suppression	Surveillance & Reference Streams	Interference-Cancelled Surveillance	Division-Order Extensive Cancellation (ECA-DO) [6]
B5	Fast Cross-Ambiguity Function	Suppressed Surveillance & Purified Reference	2D Range-Doppler (RD) Map	FFT-based Decimation CAF Algorithm [12]
B6	Adaptive Detection (CFAR)	RD Map Data	Binary Detection Hit List	Ordered-Statistic (OS) CFAR with K-Distribution [18]
B7	Density-Based Clustering	CFAR Hit Coordinates	Aggregated Target Reports	IGWOA-Optimized DBSCAN [21]
B8	Non-Linear Tracking	Target Reports & Metadata	State Estimates (Position/Velocity)	UKF-PDA with 3D Ellipsoidal Intersection [10]
B9	Multistatic Fusion	Single-node Track Data	Unified Situational Awareness	Centralized Track-to-Track Fusion [13]

Block B1: Phase-Coherent RF Front-End

The architectural foundation of a perfect passive radar resides in its hardware front-end. Because the target echoes are typically 60 to 100 dB weaker than the direct-path illumination signal, the receiver must possess extraordinary dynamic range and linearity to prevent the front-end from being saturated by the direct transmitter signal.[1]

A standard implementation utilizes a circular uniform array (CUA), often with eight or nine dipole elements.[4] This configuration is chosen for its ability to provide 360-degree azimuthal coverage and its suitability for spatial processing.[14] The interconnects from the antenna elements feed into low-noise amplifiers (LNAs) and then into high-bit-depth analog-to-digital converters (ADCs), typically 14 or 16 bits, to preserve the dynamic range necessary for

subsequent digital cancellation.[2] The output of this block is a set of time-synchronized, phase-coherent I/Q (In-phase and Quadrature) baseband streams.[1]

Block B2: Adaptive Digital Beamforming and Spatial Filtering

Once digitized, the multi-element signals enter the spatial processing stage. The goal is two-fold: to maximize the signal-to-noise ratio (SNR) of the reference signal by steering a beam toward the transmitter, and to minimize the interference in the surveillance channel by placing spatial nulls in the direction of the transmitter and any known ground-based clutter sources.[2]

The optimal processing technique here is adaptive digital beamforming. Unlike fixed beamforming, adaptive DBF calculates weights dynamically based on the received signal environment.[2] Methods like Maximal Ratio Combining (MRC) can be used when the interference is separable, maximizing the power of the desired reflection.[8] However, in complex environments, more sophisticated null-steering is required to protect the receiver's limited dynamic range before the signal reaches the heavy computational blocks of clutter cancellation.[2]

Block B3: Optimal Reference Signal Reconstruction

A significant leap in PCL performance is achieved through signal reconstruction. In the reference channel, the receiver captures the direct signal from the transmitter.[8] However, this captured signal is inevitably corrupted by noise and local multipath.[16] Using this "dirty" reference for cross-correlation limits the system's sensitivity to the noise floor of the reference channel.[16]

For digital signals like DVB-T or DTMB (Digital Terrestrial Multimedia Broadcasting), the optimal processing involves a full demodulation-remodulation loop.[15] The system demodulates the signal to its original bitstream, applies Forward Error Correction (FEC) to eliminate bit errors, and then remodulates a "perfect" digital replica of the transmitted waveform.[16]

Furthermore, advanced techniques use Singular Value Decomposition (SVD) for noise reduction in specific digital structures, such as the pseudo-noise (PN) sequences used in DTMB.[16] By exploiting the low-rank structure of these sequences, the system can significantly improve the quality of the reference signal, leading to sharper peaks in the ambiguity function.[16] This "perfect" reference replica is then passed to both the cancellation block and the cross-ambiguity engine.[17]

The Core Interference Cancellation Engine: Block B4

The single greatest bottleneck in passive radar performance is the suppression of the Direct Path Interference (DPI) and Multipath Interference (MPI).[1] These stationary or slowly varying

components are orders of magnitude stronger than the target echoes and must be removed to reveal targets submerged in their sidelobes.[6]

Extensive Cancellation Algorithm (ECA) and its Variants

The Extensive Cancellation Algorithm (ECA) is the gold standard for PCL clutter mitigation.[6] Unlike adaptive filters like the Least Mean Squares (LMS), which are iterative and may suffer from slow convergence or instability, the ECA is an open-loop algorithm based on the least squares (LS) principle.[6]

The ECA projects the received surveillance signal onto a subspace that is orthogonal to the direct signal and its various delayed versions (multipath).[6] While the standard ECA is computationally intensive due to the requirement of high-dimensional matrix inversions, the optimal implementation utilizes the **Division-Order ECA (ECA-DO)**.[6] ECA-DO divides the delay matrix into smaller sub-matrices, enabling highly parallel processing on modern hardware like FPGAs or GPUs, which is essential for the real-time operation of a "perfect" radar system.[6]

Algorithm	Processing Type	Convergence	Complexity	Hardware Suitability
LMS	Iterative/Closed-Loop	Slow/Unstable [25]	Low	Moderate
RLS	Iterative/Closed-Loop	Fast	High	Low
ECA	Least Squares/Open-Loop	Instantaneous [6]	High (O)	Moderate
ECA-B	Batch Open-Loop	Instantaneous	Medium	High (Parallel) [26]
ECA-DO	Division-Order LS	Instantaneous	Optimized/Lower [6]	Excellent (FPGAs) [6]

The output of Block B4 is a "suppressed" surveillance signal where the zero-Doppler clutter has been removed, but the Doppler-shifted target echoes remain intact. This signal is the

primary input to the detection engine.[2]

Block B5: High-Efficiency Cross-Ambiguity Processing

The detection of targets in PCL is performed by evaluating the Cross-Ambiguity Function (CAF), which correlates the suppressed surveillance signal with the purified reference signal across a range of potential delays (τ) and Doppler shifts (f).[12] This produces the 2D Range-Doppler (RD) map.

The Fast-CAF Algorithm

Calculating the CAF via "brute force" is computationally prohibitive for real-time systems.[12] The optimal implementation uses an FFT-based approach combined with decimation.[12] By recognizing that the Doppler shifts of interest (e.g., < 500 Hz for aircraft) are significantly smaller than the signal bandwidth (e.g., 8 MHz for DVB-T), the signals can be decimated after an initial block-wise correlation.[12]

The fast CAF algorithm partitions the total processing interval into M blocks of L samples.[12] The optimal processing follows this mathematical approximation:

$$\chi(\tau, f) \approx \sum_{k=0}^{M-1} g(k, \tau) \exp\left(\frac{-j2\pi k M f}{F_s}\right)$$

$$g(k, \tau) = \sum_{l=0}^{L-1} s_{clean}(l + kM) r^*_{pure}(l + kM - \tau)$$

This structure allows the system to use the Fast Fourier Transform (FFT) across the M blocks, drastically reducing the number of complex multiplications.[12] The result is a high-resolution RD map where targets appear as distinct power spikes against the residual noise floor.[12]

Adaptive Detection and Clustering: Blocks B6 and B7

Target detection in the RD map requires a thresholding mechanism that adapts to the local noise and interference environment. This is the role of Constant False Alarm Rate (CFAR) processing.[19]

Optimal CFAR for Non-Gaussian Clutter

While Cell-Averaging CFAR (CA-CFAR) is optimal for homogeneous Rayleigh noise, the residual background in a PCL system is often non-Gaussian and spiky, especially in maritime

or urban environments.[18] The "perfect" system employs **Ordered-Statistic CFAR (OS-CFAR)**.[19] OS-CFAR sorts the values in the reference window and selects a specific rank (e.g., the median) to estimate the noise floor. This makes the detector robust against "masking," where secondary targets or interference spikes in the reference window would otherwise artificially raise the threshold and cause the system to miss valid targets.[19]

Furthermore, in environments where sea clutter dominates, the noise is modeled using a **K-distribution**.[18] Optimal processing in this regime uses a square-law detector and an autoregressive technique to estimate the "texture" of the clutter, which significantly improves detection probability for low-RCS targets compared to traditional CA-CFAR.[18]

Target Extraction via DBSCAN Clustering

The output of the CFAR detector is a list of "hit" coordinates in the RD map.[28] Because a single target reflection typically spans multiple range and Doppler cells, these hits must be grouped.[21]

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is the optimal processing choice for clustering in radar maps.[21] Unlike K-means, which requires the number of clusters to be predefined, DBSCAN automatically identifies the number of targets based on the density of the hits.[21] It is particularly effective at handling the irregular, non-spherical shapes of radar detections and robustly identifying (and discarding) isolated noise hits as outliers.[21] Modern implementations use optimization algorithms, such as the Whale Optimization Algorithm (WOA), to automatically tune the DBSCAN parameters (ϵ and $MinPts$) in real-time.[22]

Block B8: Non-Linear State Estimation and Tracking

The final high-level stage in the PCL pipeline is tracking. This block converts the bistatic range and Doppler measurements into 3D Cartesian coordinates (latitude, longitude, altitude) and velocity vectors.[10]

The Primacy of the Unscented Kalman Filter (UKF)

Tracking in PCL is inherently non-linear. The relationship between the target's position and the observed bistatic range is ellipsoidal, and the EKF (Extended Kalman Filter), which relies on first-order linearization, often produces biased and unstable estimates in these scenarios.[10]

The **Unscented Kalman Filter (UKF)** is the optimal choice for passive radar tracking.[10] The UKF uses the unscented transform to propagate a set of deterministic "sigma points" through the actual non-linear equations, capturing the mean and covariance with much higher accuracy than the EKF.[10] When integrated with **Probabilistic Data Association (PDA)**, the

UKF-PDA tracker can effectively manage the uncertainty of measurement origin in cluttered environments, providing consistent and reliable target trajectories.[10]

Filter Type	Linearization Method	Accuracy	Stability	Computational Cost
EKF	Taylor Expansion (Jacobian)	Low/Biased	Moderate	Low [33]
UKF	Unscented Transform	High	High	Moderate [10]
Particle Filter	Monte Carlo Sampling	Excellent	High	Prohibitive [10]

Block B9: Multistatic Fusion and Networking

A standalone PCL receiver is limited by the geometry of a single transmitter-receiver pair. To achieve high-precision 3D localization, the system must fuse data from multiple nodes or multiple transmitters.[4]

In a centralized fusion architecture, the individual target reports (ellipsoids) from each node are transmitted to a Central Node (CN).[13] The optimal processing at the CN involves intersecting these ellipsoids to solve for the target's state.[4] Emerging paradigms like **Cooperative Passive Coherent Location (CPCL)** leverage the existing infrastructure of 5G mobile networks, turning the network itself into a distributed MIMO radar.[5] This allows for joint optimization of radio resources and radar performance, enabling a level of situational awareness that far exceeds the capabilities of traditional active radar networks.[5]

Computational Implementation and Real-Time Constraints

The "perfect" passive radar is not just an algorithmic ideal but a hardware challenge. The throughput required for real-time CAF and ECA processing often exceeds 100 Mbps per channel, necessitating a highly pipelined architecture.[1]

Modern systems prioritize **pipelining and parallelism** to minimize latency.[1] Feedback loops in algorithms are avoided wherever possible to allow the hardware to process data in a continuous stream.[1] Memory access latency is another critical bottleneck; thus, the optimal

architecture ensures that frequently accessed data—such as the reference signal blocks—are kept in local L1/L2 caches or embedded FPGA memory to maximize the execution speed of the FFT and cancellation units.[1]

The integration of software-defined radio (SDR) and mobile edge computing (MEC) facilities, as planned for 5G, will allow these sophisticated processing pipelines to be deployed ubiquitously, fulfilling the vision of a silent, invisible, and all-seeing radar network.[5] By meticulously optimizing each block—from SVD-based reference purification to UKF-PDA tracking—the passive radar system achieves the performance parity required for high-stakes defense and security applications.[5]

Works cited

1. Efficient Signal Processing Algorithms for Passive Radars - NATO, accessed February 9, 2026, https://publications.sto.nato.int/publications/STO%20Meeting%20Proceedings/STO-MP-SET-187/MP-SET-187-12.pdf
2. Passive radar - Wikipedia, accessed February 9, 2026, https://en.wikipedia.org/wiki/Passive_radar
3. Method for Real-Time Signal Selection for Passive Coherent Location Systems - ucf stars, accessed February 9, 2026, https://stars.library.ucf.edu/etd/5921/
4. Passive Coherent Location Radar - The Silent Threat - DSIAC, accessed February 9, 2026, https://dsiac.dtic.mil/articles/passive-coherent-location-radar-the-silent-threat/
5. Cooperative Passive Coherent Location: A Promising 5G Service to Support Road Safety - arXiv, accessed February 9, 2026, https://arxiv.org/pdf/1802.04041
6. An Improved Extensive Cancellation Method for Clutter Removal in ..., accessed February 9, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12610278/
7. Improved LMS algorithm and its application in direct path and multipath cancellation for passive radar | Request PDF - ResearchGate, accessed February 9, 2026, https://www.researchgate.net/publication/261524366_Improved_LMS_algorithm_and_its_application_in_direct_path_and_multipath_cancellation_for_passive_radar
8. Beamforming Techniques for Passive Radar: An Overview - MDPI, accessed February 9, 2026, https://www.mdpi.com/1424-8220/23/7/3435
9. Passive Bistatic Radar Performance Assessment - MATLAB & Simulink - MathWorks, accessed February 9, 2026, https://www.mathworks.com/help/radar/ug/passive-bistatic-radar-performance-assessment.html
10. Passive Radar Tracking in Clutter Using Range and Range-Rate ..., accessed February 9, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10300973/
11. Beamforming Techniques for Passive Radar: An Overview - PMC, accessed February 9, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10098532/
12. Passive radar signal processing part 2: range – doppler processing ..., accessed February 9, 2026,

https://dopplerfish.com/passive-radar-signal-processing-part-2/
13. Statistical Properties of Target Localization Using Passive Radar Systems The research was funded by the strategic innovation programme “Smartare Elektroniksystem”, a joint research project financed by VINNOVA, Formas and the Swedish Energy Agency, and by SAAB. - arXiv, accessed February 9, 2026, https://arxiv.org/html/2601.20817v1
14. Block diagram processing signal for a passive radar based on array circular uniform., accessed February 9, 2026, https://www.researchgate.net/figure/Block-diagram-processing-signal-for-a-passive-radar-based-on-array-circular-uniform_fig2_333430317
15. Signal Processing for Passive Radar Using OFDM Waveforms - Carnegie Mellon University, accessed February 9, 2026, https://users.ece.cmu.edu/~crberger/J-STSP-MRA-00514-2009-final.pdf
16. A Joint Low-Rank and Sparse Method for Reference Signal ..., accessed February 9, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8196861/
17. DVB-T Passive Radar Signal Processing, accessed February 9, 2026, https://jenda.hrach.eu/f2/passive-radar-processing-preprint.pdf
18. Optimal Detection in the K-Distributed Clutter Environment -- Non-Coherent Radar Processing - Defence Science and Technology Group, accessed February 9, 2026, https://www.dst.defence.gov.au/sites/default/files/publications/documents/DSTO-TR-2785.pdf
19. Constant False-Alarm Rate (CFAR) - Radartutorial.eu, accessed February 9, 2026, https://www.radartutorial.eu/01.basics/False%20Alarm%20Rate.en.html
20. FPGA Implementation of Efficient CFAR Algorithm for Radar Systems - PMC - NIH, accessed February 9, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9861839/
21. DBSCAN vs. K-Means: A Guide in Python - New Horizons, accessed February 9, 2026, https://www.newhorizons.com/resources/blog/dbscan-vs-kmeans-a-guide-in-python
22. Improved WOA-DBSCAN Online Clustering Algorithm for Radar Signal Data Streams - MDPI, accessed February 9, 2026, https://www.mdpi.com/1424-8220/25/16/5184
23. K-Means vs. DBSCAN: Clustering Algorithms for Grouping Data | by Hassaan Idrees, accessed February 9, 2026, https://medium.com/@hassaanidrees7/k-means-vs-dbscan-clustering-algorithms-for-grouping-data-a4969034cfcc
24. Nonparametric Multitarget Data Association and Tracking for Multistatic Radars - IEEE Xplore, accessed February 9, 2026, https://ieeexplore.ieee.org/iel8/10057477/10810753/10803016.pdf
25. Joint multipath signals and noise reduction in passive radar - IET Digital Library, accessed February 9, 2026, https://digital-library.theiet.org/doi/full/10.1049/sil2.12100
26. Batch Version of Extensive Cancellation Algorithm for Clutter Mitigation in

Frequency Domain of Passive Radar - 雷达学报, accessed February 9, 2026, https://radars.ac.cn/en/article/doi/10.12000/JR15098
27. Intelligent Detection of Low–Slow–Small Targets Based on Passive Radar - MDPI, accessed February 9, 2026, https://www.mdpi.com/2072-4292/17/6/961
28. Constant False Alarm Rate (CFAR) Detection - MATLAB & Simulink - MathWorks, accessed February 9, 2026, https://www.mathworks.com/help/phased/ug/constant-false-alarm-rate-cfar-detection.html
29. Adaptive Radar Detection: Part 1 - Medium, accessed February 9, 2026, https://medium.com/@itberrios6/adaptive-radar-detection-part-1-ec3c318d5da1
30. Switching CA/OS CFAR using Neural Network for Radar Target Detection in Non-Homogeneous Environment - ResearchGate, accessed February 9, 2026, https://www.researchgate.net/publication/282787905_Switching_CAOS_CFAR_using_Neural_Network_for_Radar_Target_Detection_in_Non-Homogeneous_Environment
31. CFAR Detection in Heterogeneous K-Distributed Sea-Clutter Background - CERES Research Repository, accessed February 9, 2026, https://dspace.lib.cranfield.ac.uk/bitstreams/609845cc-91ae-4327-bf05-4c96ce798c4c/download
32. DBSCAN vs K-Means: Overcoming Clustering Limitations in Machine Learning, accessed February 9, 2026, https://blog.quantinsti.com/dbscan-vs-kmeans/
33. EKF- and UKF-Based Estimators for Radar System - Frontiers, accessed February 9, 2026, https://www.frontiersin.org/journals/signal-processing/articles/10.3389/frsip.2021.704382/full
34. Comparison of MHT and PDA track initiation performance | Request PDF - ResearchGate, accessed February 9, 2026, https://www.researchgate.net/publication/261388005_Comparison_of_MHT_and_PDA_track_initiation_performance
35. Target Tracking Using Probabilistic Data Association-Based techniques with Applications to Sonar, Radar, and EO Senso - DSP-Book, accessed February 9, 2026, https://dsp-book.narod.ru/HMDF/2379ch08.pdf
36. Multistatic passive coherent location resource optimization - Penn State Research Database, accessed February 9, 2026, https://pure.psu.edu/en/publications/multistatic-passive-coherent-location-resource-optimization/
37. Joint Optimization of Receiver Placement and Illuminator Selection for a Multiband Passive Radar Network - MDPI, accessed February 9, 2026, https://www.mdpi.com/1424-8220/17/6/1378
38. What Does Radar Data Look Like? — Radar Academy - TI Developer Zone, accessed February 9, 2026, https://dev.ti.com/tirex/explore/node?node=A__AUAJwr4zBXhyz-M3VG2cUg__RADAR-ACADEMY__GwxShWe__LATEST
39. Flow chart of radar data preprocessing. | Download Scientific Diagram - ResearchGate, accessed February 9, 2026,

https://www.researchgate.net/figure/Flow-chart-of-radar-data-preprocessing_fig5_379199321